# Microeconomics I — Practice for the Part I exam

Class 9 · Tuesday 27 October 2026 · Pavel Kocourek · Universitat de Barcelona

Derive every result by hand: the steps matter as much as the answers. In class, you try the first part of each problem in pairs, and then we derive it on the board. Worked solutions are posted on the course page after class.

**Notation.** Prices $p = (p_1, p_2) \gg 0$, wealth $w > 0$, bundles $x = (x_1, x_2) \in \mathbb{R}^2_+$. $x(p, w)$: Walrasian demand; $v(p, w)$: indirect utility; $e(p, u)$: expenditure function; $h(p, u)$: Hicksian demand. When prices change from $p^0$ to $p^1$ with $w$ fixed (the superscripts 0 and 1 mark before and after, not powers; $p_1$ is the price of good 1, $p^1$ the price vector after the change), $u^0 = v(p^0, w)$ and $u^1 = v(p^1, w)$, and the welfare measures are $\mathrm{CV} = w - e(p^1, u^0)$ (compensating variation) and $\mathrm{EV} = e(p^0, u^1) - w$ (equivalent variation), the signs of Mas-Colell, Whinston and Green (MWG): both are negative for a loss.

## Problem 1 · Cobb–Douglas: from the known demand to CV and EV

classes 4–7

A consumer has the utility function $u(x_1, x_2) = x_1\, x_2^2$.

(a) Use the Cobb–Douglas share rule (each good gets the share of wealth equal to its exponent divided by the sum of the exponents) to write her Walrasian demand $x(p, w)$. Evaluate it at $p = (1, 1)$, $w = 6$.

(b) Find the indirect utility function $v(p, w)$.

(c) Find the expenditure function $e(p, u)$ by inverting $v$.

(d) Find the Hicksian demand $h(p, u)$ by Shephard's lemma. Check that $h(p, u) = x(p, e(p, u))$ for all $p$ and $u$.

(e) The price of good 2 rises: prices move from $p^0 = (1, 1)$ to $p^1 = (1, 8)$, and $w = 6$ stays the same. Find the new bundle. Split the change in $x_2$ into the substitution effect $h_2(p^1, u^0) - x_2(p^0, w)$ and the wealth effect $x_2(p^1, w) - h_2(p^1, u^0)$. Compare good 1's Hicksian demand $h_1(p^1, u^0)$ with $x_1(p^0, w)$: what does it show? Compute CV and EV.

## Problem 2 · $u = 2\sqrt{x_1} + 3\sqrt{x_2}$: solve the utility maximization problem

classes 4–7

A consumer has the utility function $u(x_1, x_2) = 2\sqrt{x_1} + 3\sqrt{x_2}$.

(a) Set up the Lagrangian of her problem, maximize $u(x)$ subject to $p_1 x_1 + p_2 x_2 \le w$, with one multiplier for the budget constraint. Write the first-order conditions and solve them for $x(p, w)$. Explain why the optimum is interior (both goods are bought) and why the first-order conditions are sufficient here. Evaluate $x(p, w)$ at $p = (1, 3)$, $w = 56$.

(b) Find $v(p, w)$ and simplify it to a single square root.

(c) Find $e(p, u)$.

(d) Find $h(p, u)$ by Shephard's lemma.

(e) The price of good 1 rises: prices move from $p^0 = (1, 3)$ to $p^1 = (8, 3)$, and $w = 56$ stays the same. Find the new bundle and her new utility. Compute CV and EV (tip: work with the square of utility, $(u^0)^2$, rather than $u^0$). Split the change in $x_1$ into the substitution effect $h_1(p^1, u^0) - x_1(p^0, w)$ and the wealth effect $x_1(p^1, w) - h_1(p^1, u^0)$.

## Problem 3 · Expected utility: a two-stage game

class 8

Martina maximizes expected utility. Her Bernoulli utility over money is $u(x) = \sqrt{x}$, for $x \geq 0$ euros. She throws a fair die.

- On 1 or 2 she plays lottery $A$: 144 euros with probability $\frac{1}{4}$, otherwise 0 euros.
- On 3, 4, 5 or 6 she plays lottery $B$: 81 euros or 9 euros, with probability $\frac{1}{2}$ each.

(a) Write the reduced (simple) lottery of this game: each prize with its probability.

(b) Compute her expected payoff $E[x]$ in euros, her expected utility $E[u(x)]$, her certainty equivalent $c$ (the sure amount with $u(c) = E[u(x)]$) and the risk premium $\mathrm{RP} = E[x] - c$. What does the shape of $u$ say about her attitude to risk, and where does it show in these numbers?

(c) Before the die is thrown, Martina is offered $s$ euros for sure in place of lottery $B$: on 3, 4, 5 or 6 she would get $s$ euros; on 1 or 2 she still plays lottery $A$. Use the independence axiom to explain why her choice does not depend on lottery $A$. For which amounts $s$ does she weakly prefer the offer?

(d) Pau is risk neutral and faces the same game and the same offer. For which $s$ does he weakly prefer the offer? Would a strictly risk-loving person need a larger or a smaller $s$ than Pau? Why?

## Problem 4 (optional) · Properties of preferences

class 2

Joan values good 1 on its own and also uses good 2, but only together with good 1, one unit with one unit (extra units of good 2 are useless to her). Her utility function on $\mathbb{R}^2_+$ is $u(x_1, x_2) = x_1 + \min\{x_1, x_2\}$. Are her preferences

(a) monotone: $y \gg x$ implies $y \succ x$?

(b) strongly monotone: $y \geq x$ and $y \neq x$ imply $y \succ x$?

(c) locally non-satiated (LNS)?

(d) convex?

For each: *yes*, with a reason, or *no*, with a counterexample.

Here $y \gg x$ means $y_1 > x_1$ and $y_2 > x_2$; $y \geq x$ means $y_1 \geq x_1$ and $y_2 \geq x_2$ (MWG Def. 3.B.2). LNS: for every bundle $x$ and every $\varepsilon > 0$ there is a bundle $y$ within distance $\varepsilon$ of $x$ with $y \succ x$ (MWG Def. 3.B.3). Convex: for every $x$, the upper contour set $\{y \in \mathbb{R}^2_+ : y \succsim x\}$ is convex (MWG Def. 3.B.4).

Worked solutions to all four problems are posted on the course page after class.